```
% -------- 9. Travelling Salesman Problem in Prolog --------

% Distance between cities
distance(a,b,10).
distance(a,c,15).
distance(a,d,20).

distance(b,a,10).
distance(b,c,35).
distance(b,d,25).

distance(c,a,15).
distance(c,b,35).
distance(c,d,30).

distance(d,a,20).
distance(d,b,25).
distance(d,c,30).

% Calculate total cost of a route
cost(A,B,C,D,Cost):-
    distance(A,B,D1),
    distance(B,C,D2),
    distance(C,D,D3),
    distance(D,A,D4),
    Cost is D1+D2+D3+D4.

% Main Program
```

```
tsp:-

    cost(a,b,c,d,C1),
    cost(a,b,d,c,C2),
    cost(a,c,b,d,C3),
    cost(a,c,d,b,C4),
    cost(a,d,b,c,C5),
    cost(a,d,c,b,C6),

    write('Route 1 : A -> B -> C -> D -> A = '), write(C1), nl,
    write('Route 2 : A -> B -> D -> C -> A = '), write(C2), nl,
    write('Route 3 : A -> C -> B -> D -> A = '), write(C3), nl,
    write('Route 4 : A -> C -> D -> B -> A = '), write(C4), nl,
    write('Route 5 : A -> D -> B -> C -> A = '), write(C5), nl,
    write('Route 6 : A -> D -> C -> B -> A = '), write(C6), nl.
```